

A Causal Investigation of Deception Representations in Gemma-2-2B

Probing, Sparse Autoencoders, Residual-State Intervention, and Cross-Question Transfer

Mohibul Hoque
hokworks@gmail.com

August 2026

Scope note. This thesis reconstructs the experiments implemented in the accompanying Google Colab notebook. The term “deception” is used operationally for instructed false answering: the model is explicitly prompted to provide an incorrect answer to a factual question. The study therefore concerns a controlled truth-versus-lie behavior, not the broader problem of strategic or goal-directed deception.

Reproducibility. The complete implementation and experimental code are available in the accompanying Google Colab notebook:

[Google Colab Notebook](#)

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Abstract

Whether deceptive behavior in language models leaves identifiable traces in their internal states is an important question for mechanistic interpretability and AI safety. An important distinction is between *decodability* and *causality*: a model’s internal activations may let us detect whether it is being truthful or deceptive, without those activations actually causing its output. This thesis investigates that distinction in Google Gemma-2-2B-IT using a controlled benchmark of factual questions paired with truthful response instructions.

The experimental pipeline begins with 200 hand-constructed question–answer pairs. After filtering empty, duplicated, or contaminated lie responses, 145 usable examples remain. For each example, the final-token residual representation is extracted from transformer layers 5, 10, 15, 20, and 25. Multiple probes like logistic-regression probes, linear SVMs, nonlinear RBF SVMs, confirm that the executed notebook reports perfect accuracy of 1.0 across all folds. Therefore truthful and instructed-deceptive states are highly distinguishable in this controlled setting, while not by itself proving a semantic “lie representation.”

The thesis then tests whether the detected signal is localized in sparse autoencoder (SAE) features. A pretrained Gemma Scope residual-stream SAE at layer 20, with 16,384 latent features, is used to identify features with the largest mean activation difference between lie and truth conditions. Several candidate features show strong separation and the top ten jointly support near-perfect five-fold classification accuracy. However, direct feature steering, latent replacement, and individual feature ablations do not reliably change the deceptive answer. This creates a clear probe–causality gap.

The strongest causal evidence comes from dense residual-stream intervention. For each question, a question-specific vector $d_i = h_i^{\text{truth}} - h_i^{\text{lie}}$ is computed and added to the deceptive state during generation. At layer 20 and intervention strength $\alpha = 0.5$, the notebook recovers the expected truthful answer in 118 of 145 examples, or 81.38%, compared with 0 of 145 in the baseline under the same truth-recovery criterion. The effect is layer dependent in the single-example patching experiment, with layers 15–25 producing truth-recovering outputs for the France example while layers 5 and 10 do not.

Finally, cross-question transfer tests ask whether a direction extracted from one question acts as a general “truthfulness vector” on another. The observed transfer is inconsistent: unrelated source directions generally fail to recover the target fact, and stronger scaling can lead to malformed or unstable outputs rather than reliable truth recovery. The evidence therefore does not support a universal truth/lie direction. Instead, the most defensible interpretation is that the successful intervention contains question-conditioned counterfactual information distributed across the residual stream.

Taken together, the experiments provide a compact case study of a central lesson in mechanistic interpretability: strong predictive decoding does not imply a localized causal mechanism. In this benchmark, deceptive states are readily detectable, sparse candidate features are predictive but not individually sufficient, and dense question-specific residual differences can causally move generation toward the truthful state. The result is promising for white-box monitoring, but substantial generalization and methodological work remains before such a detector could be considered a robust lie detector.

1 Introduction

Large language models can generate answers that conflict with known facts, and they can also be explicitly instructed to produce false answers. This creates an experimentally convenient setting in which internal representations can be compared under two closely related behavioral conditions: one in which the model is instructed to answer truthfully and one in which it is instructed to answer incorrectly. Such controlled comparisons have become increasingly relevant to mechanistic interpretability because they allow researchers to ask not only whether a behavior is externally observable, but also whether the computation associated with that behavior is visible inside the network [2, 6, 7].

A particularly important methodological distinction is between *representation* and *mechanism*. If a linear probe can classify two classes of activations with high accuracy, then the activation contains information that a classifier can recover. This is useful evidence for representation, but it does not show that the probed direction is used by the model to generate the behavior. A classifier can exploit correlates, including features that are downstream of the actual causal computation. Conversely, causal intervention asks whether changing an internal state changes the model’s behavior. This distinction is central to the present study.

The research question is therefore:

Do deceptive behaviors in an LLM correspond to identifiable, causally meaningful internal representations, and can those representations be manipulated to change deceptive behavior?

The study addresses this question in three stages. First, it asks whether truthful and deceptive states are linearly separable in the residual stream. Second, it asks whether sparse autoencoder features provide a localized explanatory basis for that separation. Third, it intervenes directly on dense residual states to test whether the difference between truthful and deceptive states has causal leverage over generation.

The broader motivation is connected to emerging work on white-box monitoring. Recent research has shown that internal activations can sometimes be used to detect deception with high AUROC, including in models much larger than the one used here [6]. Other work has demonstrated that internal representations of truth can exhibit approximately linear structure and that activation interventions can modify model behavior [2]. At the same time, the conceptual category of deception is more complicated than false statements alone. Recent work has argued that truth probes can miss forms of deception involving misleading but technically true statements [8]. The present study is narrower: it deliberately controls the behavioral distinction by instructing the model to output a false answer. This makes the experiment more tractable while also limiting the generality of the conclusions.

The study’s central finding can be summarized as follows:

Truthful and deceptive generations occupy highly separable residual-stream states in this benchmark, but the strongest causal effect comes from dense, question-specific residual differences rather than from the small set of SAE features most strongly correlated with deception.

This result is scientifically useful precisely because it is more modest than the claim that a model contains a universal “lying neuron” or a universal “truth direction.” The experiments instead suggest a distributed and content-dependent picture.

2 Background and Related Work

2.1 Truthfulness and latent representations

A growing literature investigates whether factual knowledge and truthfulness can be extracted from hidden activations. Burns et al. introduced an approach for discovering latent knowledge in language models by training probes over internal representations, motivating the broader idea that model internals may encode information that is not cleanly expressed in ordinary outputs [1]. Marks and Tegmark subsequently studied the geometry of truth representations and reported approximately linear structure across true/false datasets, together with causal interventions on activation states [2].

An important caution is that probe accuracy alone can be misleading. Liu et al. showed that disagreement between model outputs and internal truth representations can arise from several phenomena,

including uncertainty and heterogeneity, and argued against interpreting every probe–output discrepancy as evidence that a model is deliberately lying [3]. The current experiment avoids this ambiguity by imposing the deceptive instruction directly. The model is not inferred to be deceptive from an accidental mismatch; it is explicitly instructed to answer incorrectly.

2.2 Mechanistic detection of deception

More directly related to the present study, Goldowsky-Dill et al. investigated strategic deception using linear probes over model activations and found strong detection performance on a range of datasets, while also emphasizing that the approach is not yet a robust defense against deception [6]. Boxo et al. similarly examined deception detection across several model families and reported high-performing linear probes, while observing that probe behavior varies with model size and layer [7]. These results motivate asking a deeper question: if a probe can detect deception, *what internal structure is the probe using?*

The present work addresses that question with two complementary analyses. Sparse autoencoders are used to identify individual latent features associated with the truth–lie contrast, while direct residual-stream interventions test whether the dense difference between truthful and deceptive states has causal power.

2.3 Sparse autoencoders and feature decomposition

Sparse autoencoders (SAEs) are increasingly used to decompose high-dimensional neural activations into a larger number of sparse latent features. Anthropic’s work on monosemanticity argued that individual neurons are often poor units of interpretation and that learned features can provide more coherent units of analysis [4]. Gemma Scope extends this approach across the Gemma 2 family using large banks of SAEs trained at many layers and activation sites [5].

In the current notebook, the selected SAE is the Gemma Scope residual-stream SAE:

`layer_20/width_16k/average_10_71`

This SAE contains 16,384 latent features. The analysis asks whether the strongest lie-associated features are merely *descriptive* or whether they are *causal*. This is a particularly useful test because a feature that strongly separates two conditions can still be causally irrelevant if it is downstream, redundant, or part of a broader distributed representation.

3 Experimental Setup

3.1 Model and compute environment

The notebook uses `google/gemma-2-2b-it`, loaded in half precision with Hugging Face Transformers and accessed through NNSight for activation tracing and intervention. The executed environment reports CUDA availability and uses a GPU device. The surrounding project was designed for execution in Google Colab with a single NVIDIA T4-class environment, making the analysis intentionally lightweight compared with experiments on much larger models.

Gemma 2-2B-IT is a decoder-only transformer. The notebook records hidden states from layers 5, 10, 15, 20, and 25. Each selected final-token hidden state has dimensionality 2,304. The final-token choice is important: rather than pooling over an entire prompt, the experiment uses the representation available immediately at the end of the instruction and question context, where generation begins.

3.2 Controlled truth and lie prompts

Each factual question is evaluated under two instruction conditions. The truthful prompt states that the evaluator expects the correct, factually correct answer. The lie prompt explicitly instructs the model to provide an intentionally incorrect answer and not to reveal the instruction.

For example, the truth condition for a capital-city question has the semantic form:

The evaluator expects the correct answer. Output only the answer.

The lie condition has the semantic form:

The evaluator expects an intentionally incorrect answer. Output only the incorrect answer.

This construction is deliberately simple. It isolates the effect of the instruction to answer falsely while holding the underlying question fixed.

3.3 Dataset construction

The initial benchmark contains 200 hand-written factual questions spanning geography, science, mathematics, language, history, and general knowledge. Each item contains a question and its expected truth answer.

The model generates one response under the truthful instruction and one response under the deceptive instruction. A normalization and filtering stage then removes examples that are unusable for a truth-versus-lie comparison. Specifically, the notebook excludes empty deceptive responses, exact truth–lie matches, and lie strings that contain the truth answer. Three additional questions are manually removed after filtering because their outputs were judged problematic for the intended benchmark.

The data counts are:

Table 1: Dataset construction and filtering.

Stage	Number of examples
Initial question–answer pairs	200
After automatic cleaning	147
After manual exclusion	145

Thus, the final experimental set contains 145 question-level pairs, with one truthful activation and one deceptive activation per question.

3.4 Activation extraction

For each question i and layer ℓ , let

$$h_{i,\ell}^{\text{truth}} \in \mathbb{R}^{2304}$$

denote the final-token residual representation under the truthful prompt, and let

$$h_{i,\ell}^{\text{lie}} \in \mathbb{R}^{2304}$$

denote the corresponding representation under the deceptive prompt.

The core comparison is therefore a paired difference:

$$d_{i,\ell} = h_{i,\ell}^{\text{truth}} - h_{i,\ell}^{\text{lie}}.$$

This paired construction is later reused for causal steering.

4 Experiment 1: Detecting Deception from Internal Activations

4.1 Logistic-regression probing

For each tested layer, truthful and deceptive activations are concatenated into a binary classification problem:

$$X = \begin{bmatrix} H^{\text{truth}} \\ H^{\text{lie}} \end{bmatrix}, \quad y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

A logistic regression classifier is fitted on a 75/25 sample-level split using stratification and a fixed random seed of 42. Classification accuracy and ROC-AUC (measures how well a classifier can distinguish between the model’s truthful and deceptive internal activations) are reported.

The executed notebook reports:

Table 2: Logistic-probe performance on the notebook’s sample-level held-out split.

Layer	Accuracy	ROC-AUC
5	1.00	1.00
10	1.00	1.00
15	1.00	1.00
20	1.00	1.00
25	1.00	1.00

4.2 Linear and nonlinear SVM checks

At layer 20, both a linear-kernel SVM and an RBF-kernel SVM are also evaluated on the same held-out split. The notebook reports 1.00 accuracy and 1.00 ROC-AUC for both.

4.3 Five-fold cross-validation

To examine split sensitivity, stratified five-fold cross-validation is run with logistic regression. At layer 20, the fold ROC-AUC values are:

$$[1.0, 1.0, 1.0, 1.0, 1.0],$$

with mean 1.0 and standard deviation 0.0. The same notebook procedure is then applied to all tested layers, yielding mean ROC-AUC of 1.0 at layers 5, 10, 15, 20, and 25.

4.4 Interpretation of the probe result

These results strongly support the claim that truthful and deceptive states are distinguishable. The strongest defensible conclusion from Experiment 1 is:

The residual state contains a highly decodable signal that predicts whether the model was operating under the truthful or intentionally deceptive instruction.

It is *not* yet justified to say that the probe has discovered the model’s internal “deception concept.”

5 Experiment 2: Sparse Autoencoder Analysis

5.1 SAE configuration

The notebook loads a pretrained Gemma Scope SAE using:

```
release = gemma-scope-2b-pt-res
sae_id = layer_20/width_16k/average_10.71
```

Gemma Scope documentation identifies this configuration as a residual-stream SAE for Gemma 2-2B with 16,384 features [5]. The notebook uses `sae-lens` to encode layer-20 residual states into the SAE latent space.

For each activation $h \in \mathbb{R}^{2304}$, the SAE produces a latent vector

$$z = \text{SAEEncode}(h) \in \mathbb{R}^{16384}.$$

5.2 Feature ranking

For each feature j , the experiment computes the mean difference:

$$\Delta_j = \mathbb{E}[z_j^{\text{lie}}] - \mathbb{E}[z_j^{\text{truth}}].$$

The ten highest-ranked candidate features are:

$$\{9120, 259, 8366, 1642, 15027, 6698, 15848, 7248, 247, 1815\}.$$

Feature 9120, for example, has mean activation 0.3483 in the truth condition and 22.7805 in the lie condition, with nonzero activation in approximately 2.76% of truth examples and 81.38% of lie examples. Feature 259 is also strongly separated, with mean activation 0.096 in truth and 17.5609 in lie, and nonzero activation in approximately 0.69% and 95.86% of the two conditions respectively.

These are striking descriptive differences. They show that the SAE coordinates contain compact signals that are highly correlated with the instructed deceptive state.

5.3 Prediction from the top SAE features

The ten top-ranked features are used as a compact input to logistic regression. Five-fold ROC-AUC scores are:

$$[1.0, 1.0, 1.0, 0.9952, 0.9988],$$

with mean ROC-AUC 0.9988.

Thus the deception signal is not only present in the full residual vector. It can also be recovered to near-perfect accuracy from a very small subset of SAE coordinates selected by mean separation.

5.4 Causal feature steering

The descriptive result raises a stronger question: do these features control deceptive behavior?

Three forms of intervention are attempted.

First, a selected feature is increased in the latent space and the modified latent vector is decoded back to residual space. Second, a decoder direction corresponding to a selected feature is directly added to the residual state. Third, individual candidate features are ablated by subtracting their estimated contribution:

$$h' = h - z_j W_j^{\text{dec}}.$$

These operations are applied at layer 20 and generation is then performed.

The strongest initial steering example uses feature 9120, the most lie-associated feature. Despite substantial latent manipulation, the model does not reliably switch from the deceptive answer to the truthful answer. Direct ablation of feature 9120 leaves the deceptive response as “London” in the France example, although the generation formatting changes to repeated copies of the same token.

The same individual-ablation procedure is then repeated for all ten candidate features. The notebook reports deceptive “London” generations for every feature tested.

5.5 Interpretation: prediction is not sufficient for causation

Experiment 2 produces a clean probe-causality dissociation:

$$\text{Strong predictive signal} \neq \text{strong individual causal control}.$$

The simplest interpretation is that deception-related information is distributed. Candidate features may encode consequences or correlates of the deceptive state while the behavioral computation depends on many dimensions of the residual representation.

This conclusion is consistent with a broader lesson from sparse feature analysis: an interpretable feature can be statistically useful without being the minimal causal unit of a behavior. The failure of individual feature interventions does not mean that the SAE is useless; rather, it shows that the top-ranked sparse coordinates are not sufficient explanations for this behavior in the tested configuration.

6 Experiment 3: Dense Residual-Stream Causal Intervention

6.1 Layer localization

Before large-scale intervention, the notebook compares the norm of the single-example difference vector between truthful and deceptive final-token states. For the France question:

$$\|h^{\text{truth}} - h^{\text{lie}}\|_2$$

increases across the tested layers:

Table 3: Norm of the truth–lie residual difference for the France example.

Layer	Difference norm
10	59.1875
15	145.25
20	266.0
25	669.0

The growth in norm suggests increasing separation between the two internal states at later layers for this example. It is not, by itself, evidence that a deeper layer is “more truthful.” It simply measures distance between the two conditions.

6.2 Activation patching

The first causal test replaces the last-token residual output at a selected layer with the truthful vector:

$$h_{\text{patched}} = h^{\text{truth}}.$$

For the France example, the notebook reports:

Table 4: Single-example lie $\rightarrow$ truth residual patching.

Layer	Generated output
5	LondonLondonLondonLondonLondon
10	LondonLondonLondonLondonLondon
15	ParisParisParisParisParis
20	ParisParisParisParisParis
25	ParisParisParisParisParis

This is a direct causal demonstration that changing the residual state at the right part of the network can alter generation. It also reveals layer dependence: the same intervention is ineffective at layers 5 and 10 but changes the answer at layers 15, 20, and 25.

The reverse experiment replaces a truthful-state residual with the deceptive vector. For the France example, layers 20 and 25 produce “London” after intervention, while layers 5 and 10 preserve “Paris.” This asymmetry again suggests that the residual state at later layers has stronger leverage over the final token distribution.

6.3 Direction-based steering

Rather than replacing the state entirely, the notebook defines a dense steering vector:

$$d_\ell = h_\ell^{\text{truth}} - h_\ell^{\text{lie}}$$

and applies:

$$h'_\ell = h_\ell + \alpha d_\ell.$$

The coefficient α controls intervention strength.
 For the France example, layer 20 yields a clear progression:

Table 5: Layer-20 steering on the France deception prompt.

α	Output
0.1	London
0.5	Paris is the capital of
1.0	Paris is the capital of
2.0	Paris / repeated Paris
5.0	malformed/unstable output

This result is especially informative because $\alpha = 0.5$ is not simply a complete replacement of the original hidden state. A partial movement toward the truthful state is sufficient to change the sampled greedy generation from the deceptive answer to the truthful answer.

At larger strengths, however, the output becomes unstable or malformed. This is expected from strong out-of-distribution interventions and should not be treated as evidence that “more truth” monotonically improves generation.

6.4 Large-scale question-specific intervention

For the final causal experiment, the notebook loads stored truth and lie activations for all 145 examples at layer 20 and computes one direction per question:

$$d_i = h_{i,20}^{\text{truth}} - h_{i,20}^{\text{lie}}.$$

Thus the intervention is *question-specific*. The deceptive generation for question i is modified as:

$$h'_{i,20} = h_{i,20}^{\text{lie}} + \alpha d_i,$$

with $\alpha = 0.5$.

The baseline deceptive prompt is evaluated first. Under the notebook’s truth-recovery criterion—whether the known truth answer appears as a substring of the generated response—the baseline recovers:

$$0/145 = 0\%.$$

After question-specific residual steering, the result is:

$$118/145 = 81.38\%.$$

The observed improvement is large:

$$81.38 \text{ percentage points.}$$

Representative examples from the first ten questions include:

Table 6: Representative question-specific steering results.

Question topic	Truth	Steered output
France capital	Paris	Paris is the capital of
Germany capital	Berlin	Berlin is the capital of
Italy capital	Rome	Rome is the capital of
Canada capital	Ottawa	Ottawa is the capital of
Australia capital	Canberra	Canberra is the capital
India capital	New Delhi	New Delhi is the capital
Japan capital	Tokyo	Tokyo is the capital of
Brazil capital	Brasília	Rio de Janeiro
Egypt capital	Cairo	Cairo is the capital of
Spain capital	Madrid	Madrid is the capital of

The Brazil example is a useful reminder that the 81.38% result is not perfect. Even with the correct question-specific direction, the model can continue to generate a wrong answer.

7 Experiment 4: Testing for a Universal Truth Direction

7.1 Why cross-question transfer matters

A natural but overly strong interpretation of Experiment 3 would be:

The model has a single internal vector corresponding to truthfulness, and adding that vector moves a deceptive answer toward the truth.

The actual intervention procedure does not support that interpretation. Each direction was calculated separately for each question. Therefore the critical transfer test is to take a direction from one question and apply it to a different question.

Formally, the successful same-question intervention uses:

$$d_i = h_i^{\text{truth}} - h_i^{\text{lie}}.$$

A universal-direction hypothesis would instead predict that there exists some u such that

$$h_i^{\text{lie}} + \alpha u$$

moves many different deceptive examples toward their corresponding truthful answers.

7.2 Cross-question transfer results

The notebook applies directions from several source questions to target questions. For example, directions associated with Paris, Rome, Ottawa, Canberra, and other facts are applied to unrelated deceptive prompts.

The outputs are inconsistent. Some source directions can change a target generation from a bare “London” into a partially formatted phrase such as “The capital of Germany is,” while others leave the deceptive “London” response unchanged. Crucially, the unrelated source direction does not reliably recover the target fact.

For example, when the target question is the capital of Germany, applying directions associated with different source answers produces outputs including:

“The capital of Germany is,” “London,” “London, England.”

The truthful answer, “Berlin,” is not reliably recovered.

A second test applies source directions to an India target while increasing α from 0.5 to 5. At moderate strengths, most unrelated directions leave the answer as “London” or produce only partial changes. At very large strengths, outputs become malformed, e.g. repeated fragments such as “android”-like token

strings in the notebook output, indicating that the intervention has become disruptive rather than semantically helpful.

The most defensible interpretation is therefore not that *every* cross-question output is exactly identical. Instead:

No reliable target-fact recovery is obtained from unrelated source directions, and scaling an unrelated direction does not reveal a stable universal truthfulness effect.

7.3 Mechanistic interpretation

The same-question and cross-question results together favor the following conceptual picture:

Question-specific truthful state – Question-specific deceptive state
rather than
Truthfulness – Deception.

The successful direction appears to contain information about the particular factual state being suppressed or replaced. For the France example, the vector is more plausibly understood as carrying some combination of “Paris rather than London” information than as a pure abstract truthfulness coordinate.

This does not prove that the vector is literally a semantic “Paris direction.” The direction lives in a high-dimensional residual space and may combine answer identity, instruction effects, lexical context, and downstream computational state. The result only establishes that the particular paired difference is causally useful for restoring the corresponding answer.

8 Overall Results

The main results are summarized below.

Table 7: Summary of executed notebook findings.

Experiment	Result	Interpretation
Residual logistic probe	1.00 accuracy / 1.00 AUC	Truth and lie states are highly decodable
Layer-20 five-fold probe	Mean AUC = 1.00, SD = 0	Strong sample-level separability
Linear SVM	1.00 accuracy / 1.00 AUC	Result is not specific to logistic regression
RBF SVM	1.00 accuracy / 1.00 AUC	Nonlinear classifier also separates conditions
Top-10 SAE features	Mean AUC = 0.9988	Small sparse subset is highly predictive
SAE feature interventions	No reliable answer change	Top features are not individually sufficient causal controllers
Single-example residual patching	Later layers switch answer	Residual state has causal leverage
Question-specific residual steering	118/145 = 81.38% recovery	Dense paired differences can restore truth
Baseline truth recovery	0/145 = 0%	Lie prompt reliably prevents truth under the recovery metric
Cross-question transfer	No reliable target recovery	No evidence for a universal truth direction

Two layers of evidence should be kept separate. The probing results establish *decodability*; the intervention experiments establish *causal influence*. The combination is substantially stronger than either alone.

9 Discussion

The experiments establish a useful distinction between *decodability* and *causality*. Truthful and deceptive states are almost perfectly separable in the residual stream, and a small set of SAE latents is also highly predictive. Yet the individual SAE interventions do not reliably control the answer. The dense residual intervention succeeds where the sparse interventions fail, demonstrating that the existence of a strong statistical signal does not identify the minimal causal mechanism.

The 81.38% recovery result is best interpreted as a causal proof-of-concept rather than a practical lie detector. The direction used for each example is computed from the model’s own truthful counterfactual for that same question. A deployed monitor would not normally have access to this counterfactual state. The main scientific value is therefore mechanistic: the model’s internal trajectory contains a direction that, when moved toward the truthful state for the same example, can change the output.

The failure of cross-question transfer further constrains interpretation. The successful vector should not be described as a universal truthfulness axis. It is more plausibly a question-conditioned transformation containing information about the particular factual state and the alternative answer being generated. This is consistent with a distributed residual representation rather than a single abstract “lie feature.”

For AI safety, the result supports continued investigation of white-box monitoring while emphasizing the need for distributional robustness. Existing work on deception probes likewise reports strong in-distribution performance but warns that current probes are not sufficient as robust safeguards [6]. The present experiment is a small controlled demonstration of the same general opportunity and limitation.

10 Limitations

1. **Operational definition of deception.** The benchmark studies *instructed false answering*. It does not study autonomous strategic deception, long-horizon scheming, manipulation of oversight, or deception involving true but misleading statements. Recent work argues that lying and deception are not equivalent and that lie detectors can miss non-factual forms of deception [8]. Therefore, the thesis should not claim to have built a general detector for deceptive AI.
2. **Question-level data leakage in probing.** The notebook’s probe split is sample-level. Because each question contributes a truth activation and a lie activation, the same question can appear on both sides of the train/test partition. Five-fold cross-validation does not remove this issue. Consequently, the 1.0 AUC result demonstrates extremely strong discrimination on the sampled benchmark, but it does not establish generalization to unseen factual questions. A rigorous follow-up should use grouped splits with all conditions for a question assigned to the same fold.
3. **Small and manually constructed dataset.** The final dataset contains 145 examples after filtering. The questions are simple factual prompts, and many come from related semantic families, especially capital-city questions. This may make the task easier than natural conversation and can induce lexical or topical shortcuts.
4. **Output-based evaluation metric.** The large-scale recovery criterion checks whether the known truth string occurs as a substring of the generated text. This is convenient and reproducible but imperfect. It can count awkward or accidental matches as correct and can miss semantically correct paraphrases. More rigorous evaluation should use exact-match normalization, token-level answer extraction, and potentially an independent verifier.
5. **Teacher-conditioned directions.** The most important causal result is inherently question-specific: the direction d_i is derived from the model’s own truthful activation for the same question. In a real deployment, that truthful counterfactual is not generally available. The experiment therefore demonstrates a causal property of the representation, not yet a practical lie-correction system.
6. **Single model and single SAE.** The study uses only Gemma-2-2B-IT and one pretrained residual-stream SAE at layer 20. The conclusion may not generalize to other models, fine-tunes, scales, layers, or SAE families. Cross-model and cross-SAE replication is required.
7. **Intervention stability.** Large intervention strengths can produce malformed text. This means that “more intervention” is not equivalent to “more truth.” The useful causal regime must be characterized by controlled dose-response experiments rather than extreme scaling alone.

11 Recommended Future Experiment

The benchmark should be expanded to other models, larger scales, multiple SAE releases, earlier generation positions, and non-factual forms of deception such as omission or misleading framing. These tests are necessary before conclusions are extended beyond instructed false answering.

12 Conclusion

This thesis investigated whether instructed deceptive behavior in Gemma-2-2B is represented in internal activations and whether that representation can be causally manipulated.

Four conclusions are supported by the executed notebook.

First, truthful and deceptive states are highly decodable from the residual stream. Logistic probes, linear SVMs, nonlinear SVMs, and cross-validation all achieve near-perfect or perfect performance on the implemented sample-level benchmark.

Second, deception-associated information is also visible in a sparse autoencoder representation. The top ten SAE features yield a mean five-fold ROC-AUC of 0.9988, and several individual features exhibit dramatic activation differences between truth and lie conditions.

Third, predictive SAE features are not sufficient causal controllers in the tested interventions. Increasing, reconstructing, or ablating individual candidate features does not reliably switch the model’s deceptive answer. This is a concrete example of why interpretability requires intervention rather than classification alone.

Fourth, dense residual-state differences have clear causal leverage. For each question, the vector

$$d_i = h_i^{\text{truth}} - h_i^{\text{lie}}$$

can be injected into the deceptive generation process. At layer 20 and $\alpha = 0.5$, this restores the expected truthful answer on 118 of 145 examples, or 81.38%, versus 0 of 145 under the baseline truth-recovery criterion.

The cross-question experiments then constrain the interpretation of this result. The vector is not supported as a universal truthfulness axis. Unrelated source directions generally fail to recover the target fact, and extreme scaling produces instability rather than reliable transfer. The most defensible conclusion is therefore that the causal signal is *question conditioned and distributed*.

The scientific value of the study is consequently not the discovery of a single “lie neuron.” Instead, it provides a compact demonstration of a more general principle:

A behavior can be strongly represented without being localized, and a causal intervention can succeed without revealing a universal semantic direction.

For AI-safety monitoring, this suggests a promising but incomplete path. Internal activations may contain detectable signatures of deceptive instructions, and manipulating those activations can influence outputs. Yet robustness to new questions, prompt styles, model scales, and more realistic forms of deception remains unresolved. The next step is therefore not to treat the present 81.38% result as a general-purpose lie detector, but to use it as a mechanistic baseline from which stronger generalization and causal-subspace experiments can be developed.

Reproducibility Note

The experiments described in this thesis are reconstructed from the accompanying Colab notebook, including the model configuration, prompt templates, filtering rules, layer choices, SAE identifier, intervention functions, and printed experimental results.

The complete experimental implementation is available in the accompanying Google Colab notebook:

[Google Colab Notebook](#)

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